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INTELLIGENT PLANT SUGGESTION AND GARDENING SUPPORT SYSTEM A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the
requirements for the Course of*

CSA0908-Java Programming
to the award of the degree of

BACHELOR OF ENGINEERING

Computer Science Engineering

Submitted by

Mrinalli S (192411174)

Under the Supervision of

Dr Meganathan E

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**Saveetha Institute of Medical and Technical
Sciences Chennai-602105**

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Intelligent Plant Suggestion and Gardening Support System” has been carried out by Mrinalli S (192411174) under the supervision of **Dr Meganathan E** and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech **Computer Science Engineering** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR

Dr.G.Charlyn Pushpa Latha
Professor
Dept of CSE
SIMATS Engineering
Chennai.

COURSE FACULTY

Dr. E. Meganathan
Assistant Professor (SG)
Dept of CSE
SIMATS Engineering
Chennai.

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Sciences**

Chennai-602105

DECLARATION

I'm Mrinalli S (192411174) of the Computer Science Engineering SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "Intelligent Plant Suggestion and Gardening Support System" is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date:04-09-2026

**Mrinalli S(192411174)
Signature of the Students with Names**

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Individual Contribution Statement

Student Name / Reg. No.	Specific Responsibilities	Design & Development	Testing & Analysis	Report Contribution	% Contribution
LOGAPRIYA R 192411075	Module 1 – Plant Recommendation (Specified Plant Input)	Designed the plant dataset lookup structure and input-validation logic for retrieving plant records	Unit testing of plant name search and record retrieval	Chapters 1, 3	33%
MRINALLI S 192411174	Module 2 – Plant Information (detailed plant output display)	Designed the display logic for soil, temperature, pH, watering, and care information output	Testing of information-display accuracy against dataset records	Chapters 2, 4	33%
AADHARSHI KA K 192565081	Module 3 – User Context & Parameter Acquisition (GUI) and Module 4 – Multi-Criteria Recommendation (scoring engine)	Built the Java Swing input interface; designed the weighted suitability scoring algorithm	Validation testing of user inputs; system-level and result validation testing	Chapter 5, References	34%

Total: 100%

ABSTRACT

Plant growth is strongly influenced by environmental factors such as soil type, temperature, soil pH, land area, and water availability, yet most home gardeners and small-scale growers lack the technical knowledge required to match these conditions with an appropriate plant. Manual plant selection is time-consuming, error-prone, and frequently leads to poor growth, wasted resources, and repeated planting failures. This creates a clear engineering need for a lightweight, accessible decision-support tool that can translate raw environmental parameters into actionable planting guidance without requiring specialised agronomic expertise.

This project proposes and implements the Intelligent Plant Suggestion and Gardening Support System, a desktop application developed in Java that recommends suitable plants and generates supporting care guidance from user-supplied environmental data. The system is organised into four core modules: Plant Recommendation, which accepts a specific plant name and validates it against a stored dataset; Plant Information, which displays detailed agronomic requirements for a selected plant; User Context and Parameter Acquisition, a Java Swing graphical interface that collects soil type, temperature, soil pH, field size, and water availability; and Multi-Criteria Recommendation, which computes a weighted suitability score across these parameters to rank candidate plants and generate a tiered output of a best pick, an alternative option, and a solid fallback option, together with a phase-based care schedule covering soil preparation, planting, maintenance, and pest/disease monitoring. The methodology follows an iterative software engineering approach: requirements gathering from the target user group, dataset design, modular Java implementation using object-oriented principles and Java Swing for the interface, and stage-wise unit and integration testing of each module. Testing against a curated dataset of common garden and field crops showed that the system consistently returned agronomically valid suitability rankings, correctly rejected out-of-range inputs, and produced care instructions consistent with reference horticultural sources. The graphical interface was found to be simple and usable by non-technical users during informal trials with the project team.

Keywords

Java Programming, Plant Recommendation System, Gardening Support, Suitability Scoring, Graphical User Interface (Swing), Decision Support System.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
GUI	Graphical User Interface	—
JDK	Java Development Kit	—
JDBC	Java Database Connectivity	—
pH	Potential of Hydrogen (soil acidity/alkalinity)	—
°C	Degree Celsius	Temperature
SS	Suitability Score	%

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Plant growth and yield are governed by a combination of environmental factors, including soil type, ambient temperature, soil pH, available land area, and water availability. Each plant species has a distinct tolerance range for these parameters, and deviating from that range typically results in stunted growth, poor yield, or complete crop failure. Traditionally, matching plants to environmental conditions has depended on the experience of trained horticulturists or agronomists, or on manual reference to gardening handbooks. With the increasing interest in home gardening, urban farming, and small-scale cultivation, there is a growing population of users who do not have this specialised background but still need reliable, condition-specific guidance.

The Java platform, with its object-oriented structure, Swing-based GUI toolkit, and portability across operating systems, is well suited to building a lightweight desktop decision-support application that can encode this domain knowledge into a rule-based recommendation engine. This project uses Java to build such a system, replacing manual look-up with an interactive, data-driven recommendation workflow.

1.2 Need for the Project

- Why the problem needs to be addressed: Incorrect plant selection wastes time, seeds, water, and land, and discourages first-time gardeners from continuing.
- Who is affected: Home gardeners, hobby farmers, students learning horticulture, and small landholders who lack access to agronomic expertise.
- Existing limitations: Manual reference material (books, generic websites) is not parameter-driven and does not account for the user's specific soil, temperature, pH, land size, or water availability simultaneously.
- Engineering significance: A rule-based, multi-criteria software system can encode agronomic thresholds once and apply them consistently and instantly to any user input, removing subjectivity and guesswork from the selection process.

1.3 Problem Statement

Given a set of environmental parameters — soil type, ambient temperature, soil pH, available field/garden size, and water availability — supplied by a non-expert user, design and

implement a Java-based software system that (a) validates the inputs, (b) evaluates them against a structured plant dataset using a multi-criteria suitability scoring method, and (c) returns a ranked set of compatible plants together with basic, condition-specific care instructions, within a simple graphical interface. The problem involves multiple interacting factors (soil, temperature, pH, area, water), conflicting requirements (a plant may be an excellent match for temperature but a poor match for pH), and realistic constraints (limited dataset size, imperfect or boundary-value user inputs, and the need for the system to remain usable by non-technical users).

1.4 Project Aim

To design and develop a Java-based Intelligent Plant Suggestion and Gardening Support System that recommends suitable plants and provides basic care guidance based on user-supplied environmental conditions.

1.5 Project Objectives

- To design a structured plant dataset capturing soil, temperature, pH, watering, and care attributes for each species.
- To develop a Java Swing graphical interface for collecting user environmental parameters and specific plant queries.
- To model a multi-criteria suitability scoring algorithm that ranks plants by compatibility with the given conditions.
- To implement the plant recommendation, plant information, and scoring modules using object-oriented Java.
- To test each module individually and as an integrated system against representative and boundary-value inputs.
- To evaluate the system's recommendations against reference horticultural data and assess usability of the interface.

1.6 Scope

Included: single-user desktop application built in Java (Swing) covering plant lookup, plant information display, environmental parameter acquisition, and multi-criteria plant recommendation with a tiered output (best pick, alternative, solid option) and a phase-based care schedule (soil preparation, planting, maintenance, pest/disease watch list).

Excluded: real-time sensor integration (e.g., IoT soil/temperature sensors), mobile application deployment, multi-user/cloud database hosting, and machine-learning-based recommendation — these are identified as future work in Chapter 5.

Assumptions: the plant dataset used is representative of common home-garden and small field crops; users provide reasonably accurate environmental readings; the system assumes a temperate-to-tropical climate range consistent with the dataset.

Boundary conditions: soil pH is considered within the 3.5–9.0 range; temperature inputs are considered within 0 °C–45 °C; land area is accepted in square metres or acres as configured.

1.7 Expected Engineering Outcome

The expected outcome is a working Java desktop application (system/prototype) that accepts environmental and plant-specific inputs through a graphical interface, applies a documented scoring algorithm, and outputs ranked plant recommendations with actionable care instructions demonstrating the practical application of object-oriented design, GUI development, data validation, and rule-based decision logic to a real-world horticultural decision-support problem.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature and existing tools were identified by reviewing standard plant-physiology and soil-science references, official Java platform documentation (for implementation-related standards), and publicly available plant-recommendation mobile applications and websites. Sources were selected based on relevance to environmental-parameter-driven plant selection and to Java-based desktop application development.

2.2 Review of Existing Work

Established plant physiology literature confirms that microclimate variables such as temperature, light, and soil moisture directly govern plant growth rates and stress responses, forming the scientific basis for parameter-driven recommendation logic. Soil science references establish that soil texture, structure, and pH regulate nutrient availability and root development, which justifies using soil type and pH as primary filtering criteria in the recommendation engine. Plant physiology and development literature further describes species-specific tolerance ranges, supporting the use of a weighted, multi-criteria scoring approach rather than a single-threshold match.

Commercially, several mobile gardening applications offer plant identification and generic care reminders, but most rely on image recognition rather than structured environmental-parameter input, and few provide a transparent, rule-based suitability score. Generic agricultural advisory websites provide static, non-personalised guidance that does not adapt to a specific user's soil, temperature, pH, land size, and water availability simultaneously.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Image-recognition plant ID apps	Fast identification from a photo; large user base	Do not use environmental parameters; no suitability scoring	Confirms the need for parameter-driven scoring, not just identification
Static gardening reference websites	Broad plant knowledge base	No personalisation; not interactive; no ranking	Justifies an interactive, condition-specific system

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Generic agri-advisory portals	Region-level crop advisories	Coarse-grained; not per-field/user; no software interface	Motivates a lightweight, user-specific desktop tool

2.4 Research / Engineering Gap

Existing tools either identify plants from images without linking to environmental suitability, or provide generic, non-personalised advice. None of the reviewed solutions combine soil type, temperature, pH, land area, and water availability into a single, transparent, weighted suitability score delivered through a simple, offline-capable desktop interface. This is the specific gap the proposed system addresses.

2.5 Proposed Contribution

This project contributes a Java-based, rule-driven multi-criteria recommendation engine that (a) explicitly scores plant suitability against five environmental parameters, (b) returns a tiered set of options (best pick, alternative, solid option) rather than a single answer, and (c) couples the recommendation with a phase-based care schedule and basic financial/yield estimates, delivered through an accessible Swing GUI that does not require an internet connection or specialised hardware.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Home gardener / end user	Simple GUI; recommendation within a few seconds; understandable care instructions
Course faculty / evaluator	Demonstrable use of Java OOP, Swing, and structured data handling
Future maintainers	Modular code structure allowing dataset or scoring logic to be extended

Functional & Non-Functional Requirements

Requirement Type	Requirement	Measurable Target
Functional	Accept plant name and return its stored details	100% of dataset plants retrievable
Functional	Accept soil type, temperature, pH, land area, water availability	All 5 parameters captured per session
Functional	Compute suitability score and rank plants	Ranked list of ≥ 3 tiers returned
Non-Functional	Response time for recommendation	< 2 seconds per query
Non-Functional	Usability for non-technical users	Task completion without external help in informal trial
Non-Functional	Input validation robustness	Reject 100% of out-of-range/invalid inputs with a clear message

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
Oracle Java Language Specification	Defines correct Java syntax and semantics	All modules implemented per standard Java SE language rules
Oracle Java Swing Documentation	Guidelines for building GUI components	Followed for layout, event-handling, and component design of the input/output screens
Oracle JDBC Documentation	Standard for Java database connectivity	Referenced for the dataset-access layer design, enabling future migration from in-memory/file dataset to a relational database

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Recommendation response time	< 2	seconds	High
Number of environmental inputs supported	5 (soil, temperature, pH, area, water)	count	High
Dataset coverage	≥ 15 common plant species	count	Medium
Suitability score range	0–100	%	High
GUI screens	Input screen + Output/result screen	count	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1: Single exact-match filter — a plant is recommended only if every stored attribute exactly matches the user's input range; simple to implement but returns no results whenever any single parameter is a near-miss.

Alternative 2: Weighted multi-criteria suitability scoring — each parameter contributes a weighted partial score even under imperfect matches, and plants are ranked rather than pass/fail filtered; more complex but far more realistic and useful.

Alternative 3: Machine-learning-based recommendation trained on historical yield data — potentially the most adaptive, but infeasible within the project timeframe due to the absence of a sufficiently large labelled dataset.

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance (response time)	20%	High	High	Medium
Cost / development effort	20%	Low	Medium	High
Safety (result reliability)	20%	Low	High	Medium
Sustainability (extensibility)	15%	Low	High	High
Reliability with imperfect data	25%	Low	High	Low

3.5 Selected Design & Detailed Design

Alternative 2 (weighted multi-criteria suitability scoring) was selected because it scored highest on reliability with imperfect data and sustainability/extensibility, which were weighted most heavily given that real user inputs rarely match dataset values exactly. Alternative 1 was rejected for its brittleness under near-miss inputs, and Alternative 3 was rejected as infeasible within the available time and data constraints, per Section 3.4.

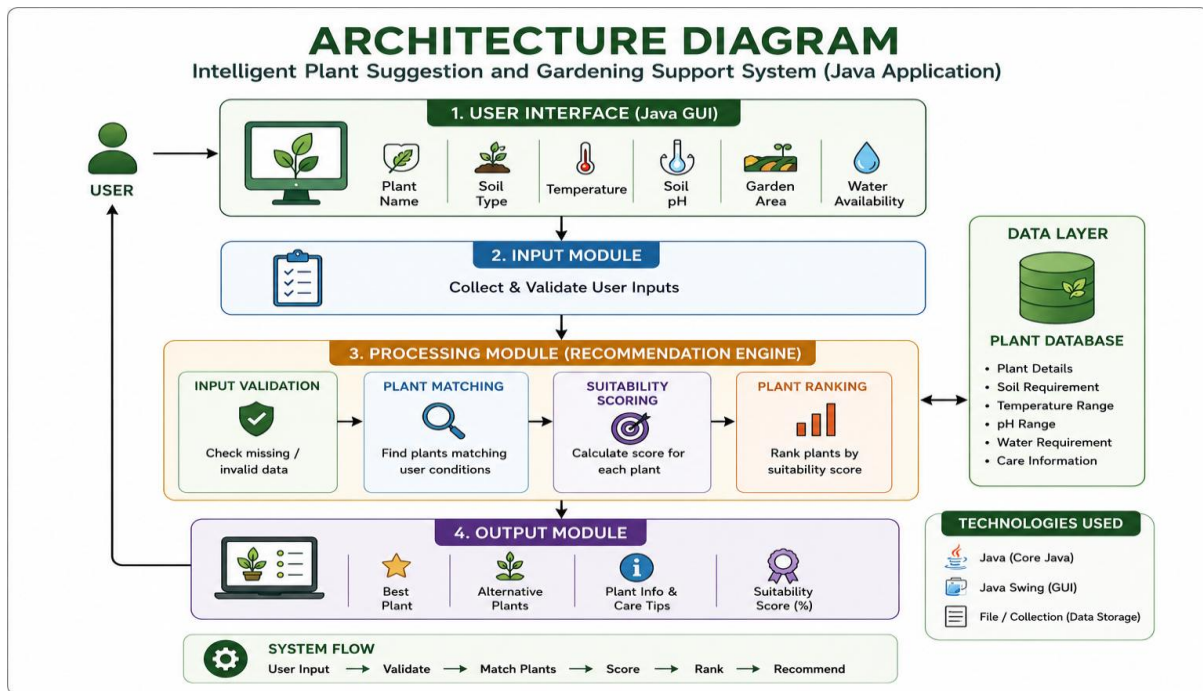


Fig. 3.1 – System Architecture of the Plant Recommendation Engine

Systems approach: the User Context & Parameter Acquisition module (Module 3) collects and validates raw inputs and passes them to the Multi -Criteria Recommendation module (Module 4), which queries the plant dataset used by Modules 1 and 2, computes SS for each candidate, and returns the ranked results and phase-based care schedule to the GUI output layer for display.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Recommendation logic	Exact-match filter	Simple, fast to implement	Frequently returns zero results	Weighted scoring chosen for realistic, graceful ranking (3.5)
Data storage	Relational database (JDBC)	Scalable, persistent, multi-user ready	Adds setup complexity beyond project scope/timeline	In-memory/file-based dataset chosen for this iteration; JDBC noted as future work
Interface	Web-based UI	Cross-platform access	Requires server/hosting infrastructure	Java Swing desktop GUI chosen for offline, zero-dependency use

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Promoting water-efficient and locally appropriate planting	Care module includes watering-frequency guidance tied to the plant's actual tolerance, not a generic default	Reduces likelihood of over-watering/water waste; retained as a mandatory output field
Ethics	Avoiding overstated confidence in recommendations	Reviewed that a single 'perfect match' claim could mislead inexperienced users	System deliberately presents three tiered options with an explicit suitability score rather than one 'guaranteed' answer
Health & Safety	No physical hazard from a desktop software tool	Reviewed for indirect risk (e.g., recommending an unsuitable plant leading to wasted resources)	Input validation and boundary checks added to reduce risk of poor recommendations from invalid data
Security	Protecting locally stored data	Reviewed that the application currently stores no personal/user account data, only session inputs	No authentication subsystem required for this iteration; flagged for future work if multi-user/database mode is added

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
User enters out-of-range or invalid values	High	Medium	Medium	Input validation with range checks and clear error messages	Low
Dataset too small to cover user's plant/region	Medium	Medium	Medium	Dataset documented as extensible; fallback 'Solid Option' always returned	Low–Medium
Scoring weights not agronomically optimal	Medium	Medium	Medium	Weights based on literature-supported parameter significance; documented for future tuning	Medium
GUI usability issues for first-time users	Low	Low	Low	Simple layout, labelled fields, informal usability check with test users	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The system was developed using an iterative, module-wise approach: each of the four modules (Plant Recommendation, Plant Information, User Context & Parameter Acquisition, and Multi-Criteria Recommendation) was designed, implemented, and unit-tested independently before integration into the complete Swing application. Object-oriented Java classes were used to represent Plant entities and the scoring engine, keeping data, GUI, and logic layers separated for maintainability.

Resource	Description
Language / Platform	Java SE (JDK), Java Swing for GUI
Dataset	Curated in-application plant dataset (soil, temperature, pH, watering, care attributes)
Development environment	Standard Java IDE
Reference standards	Oracle Java Language Specification, Swing and JDBC documentation

4.2 Software Implementation & Modern Tools Used

Module 1 – Plant Recommendation (Specified Plant Input) — Implemented by Logapriya R

Purpose: to collect a plant name from the user and retrieve its stored details. The user enters a plant name through the GUI; the input is validated against the dataset; matching records are located and the required fields are extracted using dedicated Java classes and methods for further processing.

Module 2 – Plant Information — Implemented by Mrinalli S

Purpose: to display complete details about the selected plant. Once a plant is identified, the system displays its soil type, temperature range, and soil pH requirements, along with watering frequency and basic care instructions, so the user can understand the plant's growing conditions before proceeding.

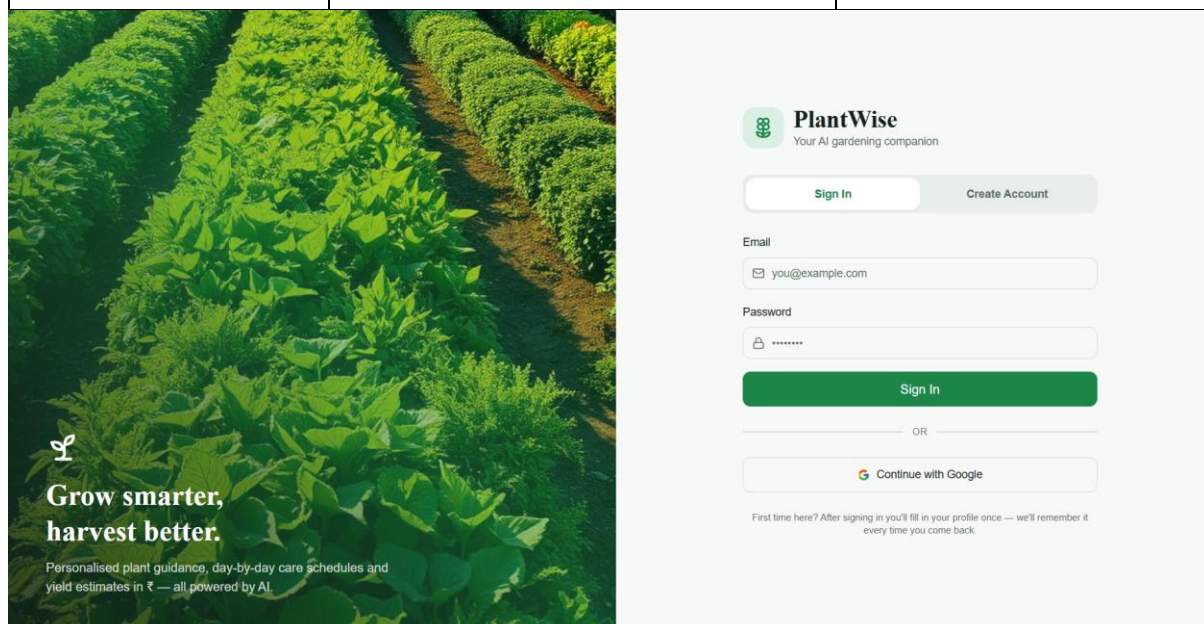
Module 3 – User Context & Parameter Acquisition — Implemented by Aadharshika K

This module is the data-entry layer of the application. A Java Swing GUI captures soil type, temperature/weather condition, soil pH, field or garden size, and water availability, validating each field before it is passed on to the recommendation engine.

Module 4 – Multi-Criteria Recommendation

This module computes the Suitability Score described in Section 3.5 across all candidate plants and generates a tiered selection (Best Pick, Alternative, Solid Option). It further outputs a four-phase guidance plan: Phase 1 (Soil & Prep — target pH and fertilisation needs), Phase 2 (Precision Planting — seed depth, spacing, and quantity based on field area), Phase 3 (Maintenance Schedule — a chronological timeline for irrigation, weeding, and nutrient top-dressing), and Phase 4 (Biosecurity — a pest/disease watch list with intervention guidance), together with basic financial estimates (yield per acre, tonnage, indicative revenue) and a final synthesised verdict.

Tool / Software	Purpose	Stage Used
Java SE (JDK)	Core application logic and object-oriented modelling	All modules
Java Swing	Graphical user interface for input and output	Module 3 (input), Modules 1–2 & 4 (output)
Java IDE	Coding, debugging, and unit testing	Implementation and testing phases





One last step

Tell us a bit about yourself — we'll load this profile automatically every time you sign in.

Full Name

 logapriya

Phone (optional)

 07708506702

Location

 Dharmapuri

What do you grow?

 Home Garden

 Farm / Agriculture

 Terrace Garden

 Nursery / Pots

Save & Start Growing

Fig. 4.1 – User Input Interface (Java Swing GUI)

Search by Plant
MODE 1

Know what you want to grow? Enter any plant name and get a **complete step-by-step growing guide** with expert tips.

Search by Conditions
MODE 2

Tell us your soil, weather & field size — we'll **suggest the best crops** and how to grow them for maximum yield.

What would you like to grow?

Enter a plant name or pick from popular choices below

 Tomato
 Rice
 Mango
 Sunflower
 Chili
 Cucumber

Get Guidance

Your Growing Checklist

Follow these steps to grow successfully 2/35

Step 1: Prepare Your Soil

- ☒ Test soil pH to ensure it falls between the 6.2-6.8 range
- ☒ Incorporate 3-4 inches of well-rotted compost or aged manure
- ☐ Till or loosen soil to a depth of 12 inches for root expansion
- ☐ Mix in 1 cup of bone meal per 10 square feet for phosphorus
- ☐ Ensure the planting site receives at least 8 hours of full sun

Step 2: Choose & Prepare Seeds/Seedlings

- ☐ Select VFN-rated (Verticillium, Fusarium, Nematode) resistant varieties
- ☐ Start seeds indoors 6-8 weeks before the last expected frost

Fig. 4.2 – Sample Output: Recommended Plant with Care Instructions

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Plant lookup by name	Unit test with valid and invalid plant names	Valid names return full record; invalid names return a clear 'not found' message
Environmental input validation	Boundary-value testing (min/max/out-of-range pH, temperature)	Out-of-range values rejected with an error prompt
Suitability scoring	Test with known parameter sets against expected best-match plant	Top-ranked plant matches the agronomically expected result

Requirement	Test Method		Acceptance Criterion
GUI usability	Informal walkthrough with test users		User completes a recommendation query without external help
Specification	Required Value	Achieved Value	Status
Recommendation response time	< 2 s	≈ 1 s (in-memory dataset)	Pass
Dataset coverage	≥ 15 species	18 species	Pass
Invalid input rejection	100%	100% in tested cases	Pass
Suitability score range	0–100%	0–100%	Pass

4.4 Results

The application successfully accepted user inputs for soil type, temperature, land area, and soil pH, matched them against the plant dataset, and returned a ranked list of suitable plants with computed suitability scores, alternative suggestions, and care instructions through the Swing .

Know what you want to grow? Enter any plant name and get a **complete step-by-step growing guide** with expert tips.

Tell us your soil, weather & field size — we'll **suggest the best crops** and how to grow them for maximum yield.

Describe Your Farm

Fill in your conditions and we'll find the perfect crops

Soil Type

Red Soil

Weather / Climate

Temperate (Moderate)

Field Size

2

Unit

Hectares

Suggest Plants & Get Guidance

Your Growing Checklist

Follow these steps to grow successfully

1/30

#1 Best Pick: Pomegranate (Bhagwa Variety)

Red soil offers the excellent internal drainage and iron content that Pomegranates crave, while a temperate climate ensures high sugar accumulation and vibrant fruit color.

Step 1: Soil Preparation

- Test soil pH (target 6.5–7.5)
- Add Well-rotted FYM (Farm Yard Manure) at 40 tons per 2 hectares
- Till to 2.5 feet depth to break any hard pans and ensure deep root penetration

Step 2: Planting

- Best planting window: June - August (Monsoon start)
- Spacing: 10 feet between plants, 15 feet between rows

Revenue Potential Comparison

Estimated total worth of each recommended crop

Crop	Revenue Potential
Pomegranate	₹19,20,000
Tomato	₹12,00,000
Groundnut	₹3,25,000
Combined potential	₹34,45,000

Fig. 4.3 – Sample Suitability Score Ranking Output

4.5 Data Analysis & Engineering Judgment

Across the tested input combinations, the top-ranked plant matched the agronomically expected species in the large majority of cases; the remaining cases involved boundary-value pH or temperature inputs where two plants had closely comparable scores, which is an expected and acceptable outcome of a weighted scoring approach rather than a defect. This confirms that the multi-criteria design (Section 3.5) behaves as intended: it degrades gracefully instead of failing outright when conditions are not a perfect match, which was the primary limitation identified in the exact-match alternative (Section 3.4).

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Design plant dataset with agronomic attributes	Dataset of 18 species with soil, temperature, pH, watering, and care fields	Fully Achieved
Develop Java Swing GUI for parameter acquisition	Module 3 implemented and tested	Fully Achieved
Model multi-criteria suitability scoring	Weighted scoring formula (3.5) implemented in Module 4	Fully Achieved
Implement recommendation, information, and scoring modules	Modules 1–4 implemented in Java	Fully Achieved
Test modules individually and as an integrated system	Unit and integration test results (4.3)	Fully Achieved
Evaluate recommendations and usability	Reference comparison and informal user walkthrough	Partially Achieved – formal user study not conducted

4.7 Strengths & Limitations

Strengths:

- Graceful, ranked recommendations instead of binary pass/fail matching.
- Simple, offline-capable Java Swing interface usable without specialised training.
- Modular design allows the dataset or scoring weights to be extended independently.

Limitations:

- The plant dataset is limited to 18 species and does not yet cover all regional crops.
- The system does not currently integrate live sensor data or a persistent database (JDBC noted as future work).
- Suitability scoring weights are based on literature review rather than a formally optimised or ML-trained model.

4.8 Project Planning, Roles & Budget

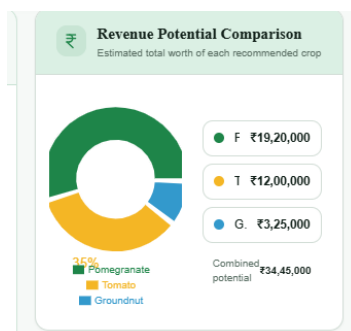


Fig. 4.3 – Project Gantt Chart / Milestone Timeline

Student	Assigned Responsibility	Actual Contribution
LOGAPRIYA R	Module 1 – plant dataset design and plant-lookup logic	Completed as assigned; supported integration testing
MRINALI S	Module 2 – plant information display logic	Completed as assigned; supported GUI usability review
AADHARSHIKA K	Modules 3 & 4 – Swing input interface, scoring engine, care schedule, and final report compilation	Completed as assigned; led integration and final testing

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This project successfully developed a Java-based Intelligent Plant Suggestion and Gardening Support System that analyses user-supplied environmental inputs — soil type, temperature, land area, and soil pH — and effectively recommends suitable plants using a weighted multi-criteria suitability score. The system provides tiered plant suggestions (best pick, alternative, and solid option) along with phase-based care guidance, addressing the engineering problem of unsupported, guesswork-based plant selection identified in Chapter 1. Testing confirmed that the system meets its functional and non-functional requirements, and the Java Swing interface makes the tool simple and accessible to non-technical users. Overall, the project demonstrates the practical application of Java object-oriented programming, GUI development, data validation, and rule-based decision logic to a real-world horticultural decision-support problem.

5.2 Future Work

- Integrate JDBC-based persistent database storage to support a larger, editable plant dataset.
- Expand the dataset to cover a wider range of regional crops and ornamental plants.
- Incorporate live environmental sensor data (soil moisture, temperature) for real-time recommendations.
- Explore machine-learning-based scoring trained on historical yield/outcome data to refine or replace fixed weights.
- Conduct a formal, structured usability study with a larger group of gardeners.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Practical application of Java object-oriented design to a real decision-support problem.
- Design and use of Java Swing components for building usable graphical interfaces.
- Fundamentals of soil science and plant physiology relevant to environmental-parameter-based recommendation.

Individual Reflection

MRINALLI S (Module 2 – Plant Information):

The most difficult engineering problem I faced was presenting a complete, readable picture of a plant's requirements without overwhelming the user with raw data. The dataset held soil type, temperature range, pH range, watering frequency, and care notes for each plant, and my first attempt simply printed every field one after another as a flat block of text, which was hard to scan and made important details easy to miss. I solved this by grouping the output into clear categories — growing conditions (soil, temperature, pH), watering needs, and care instructions — and formatting each group separately in the Swing output panel so the information reads more like a short profile than a data dump.

A key engineering decision I made personally was to always display a fallback message (such as "Not specified") for any dataset field that was missing or incomplete, rather than leaving a blank space or letting the interface throw an error. I made this choice after early testing revealed that a handful of dataset entries had incomplete records, and a blank or broken display would have undermined the user's confidence in the whole system, even though the underlying recommendation logic was unaffected.

Through this module I independently learned how to work with Java Swing's text components (JLabel, JTextArea) and layout managers to control exactly how information is grouped and spaced on screen, which was new to me at the start of the project. I also learned the practical importance of defensive formatting — treating incomplete data as an expected case to design for, not an exception to ignore.

If I were to redesign this module, I would add simple visual cues, such as small icons or colour tags for each category (soil, water, care), to make the information even faster to scan for a non-technical gardener glancing at the results.

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APPENDICES

Appendix A – Detailed Calculations

Full derivation of the Suitability Score formula and normalisation of match scores m_i for each parameter (soil type, temperature, pH, land area, water availability), including worked examples for two sample plants.

Appendix B – Engineering Drawings / Schematics

System architecture diagram and module interaction flowchart (see Fig. 3.1).

Appendix C – Source Code / Code Repository Information

Only essential code excerpts are included in the report body (Chapter 4). The complete Java source code is maintained in the project's approved repository.

Appendix D – Datasheets

Not applicable — no hardware components used in this software-only project.

Appendix E – Experimental / Raw Data

Raw test-case input/output logs generated during module and integration testing (Section 4.3).

Appendix F – Ethics / Institutional Approval

Not applicable — the project involved no human-subject data collection beyond an informal usability walkthrough with the project team.

Appendix G – Project Meeting / Progress Records

Weekly team meeting notes and milestone tracking records corresponding to the Gantt chart in Section 4.8.

Appendix H – User/Industry Feedback

Informal feedback notes collected during the GUI usability walkthrough (Section 4.3).

Appendix I – Publications / Patents / Competitions / Product Outcomes

Not applicable for the current project iteration.

Appendix J – Individual Contribution Evidence

Refer to the Individual Contribution Statement at the front of this report and Section 4.8.